

Spiral arms model documentation

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1 General considerations

This document provides a detailed and technical documentation for our model of spiral arms, implemented as a modification of IMFIT package, available at https://github.com/IVChugunov/IMFIT_spirals.

Our model describes 2D surface brightness distribution in spiral arms. Each arm is modelled independently, and model parameters correspond to arms physical properties or connected with them. The number of parameters (17 for a single arm for the most common function) makes the model flexible enough.

The model was constructed according to the results of investigation of spiral arm surface brightness profile in Chugunov et al. (2025). The key properties of the model are also described in Section 4 of this work.

Our model is represented in have 7 different functions, however all of them correspond to our model, and this implementation is done for convenience only (for example, in a function for the most common use, some of the less significant parameters are removed from the set of input parameters and set default inside).

At first, we have a a baseline function **SpiralArm** producing smooth spiral arms for the general use and describe it primarily. Next, we have two functions which only difference from the baseline that it is able to produce once- and twice-bent arms (**SpiralArm1b** and **SpiralArm2b**, respectively). The nuances of bendings are described at Section 3.1. Moreover, for each these variation we have an extension with additional parameters, allowing fine-tuning of the profile; they are called **SpiralArmFull**, **SpiralArmFull1b** and **SpiralArmFull2b** respectively, described in Section 3.2. We also have **SpiralSpur** function which is the “reduced” version of the baseline function designed to fit small elements of spiral structure.

2 Exact formula for the baseline function

To describe 2D surface brightness distribution, polar coordinates $I(r, \varphi)$ in the galaxy disc plane are used. Here, r is a galactocentric distance, φ is an azimuthal angle counted from some direction. Let the beginning of a spiral arm be located at azimuthal angle φ_0 . For a convenience we'll be use coordinate ψ (winding angle) which is counted from the beginning of a spiral arm towards its winding direction. Thus $\psi = \varphi - \varphi_0$ if spiral arm winds counterclockwise, and $\psi = \varphi_0 - \varphi$ in the opposite case. The final formula consists of a few parts:

$$I(r, \psi) = I_{\parallel}(r(\psi), \psi) \times I_{\perp}(r - r(\psi), \psi) \quad (1)$$

Here, $I_{\parallel}$ is a surface brightness distribution along the ridge-line of the spiral, which is described by the shape function of the arm $r(\psi)$. $I_{\perp}$ is a surface brightness distribution across the arm. Now let's consider each of the functions separately.

2.1 Arm shape: $r(\psi)$

Shape function of the arm $r(\psi)$ describes the overall shape of the spiral arm, more precisely defines its ridge-line. In our implementation, $\log r$ is a 4-th degree polynomial of ψ . (In the case of logarithmic spiral, $\log r$ is a linear function of ψ).

$$r(\psi) = r_0 \times \exp \left(\sum_{n=1}^4 k_n (\psi/\psi_{\text{end}})^n \right) \quad (2)$$

Coefficients k_n are not explicitly used as function parameters, but instead are derived from parameters that are more convenient for use. Parameters include the coordinates of the beginning of the arm (r_0, φ_0) , and the coordinates of its end $(r_{\text{end}}, \varphi_{\text{end}})$. These parameters also define the number of spiral revolutions and the winding direction. For example, if $\varphi_0 = 30^\circ$, $\varphi_{\text{end}} = -510^\circ$, than spiral winds clockwise and makes one and a half revolution. If $\varphi_0 = 90^\circ$, $\varphi_{\text{end}} = 450^\circ$, than spiral winds counterclockwise and makes one revolution.

For ψ coordinate, indices are used in the same way as for φ , for example ψ_{end} is a ψ coordinate where $\varphi = \varphi_{\text{end}}$.

The function also includes parameters m_2, m_3, m_4 which define the deviation of $r(\psi)$ from the logarithmic spiral.

2.1.1 k_n coefficients

k_n coefficients are derived from the following:

$$\begin{cases} k_1 = m_1 - m_2 + m_3 - m_4 \\ k_2 = m_2 - 3m_3 + 6m_4 \\ k_3 = 2m_3 - 10m_4 \\ k_4 = 5m_4 \end{cases} \quad (3)$$

where $m_1 = \ln(r_{\text{end}}/r_0)$.

2.1.2 Motivation of using $m_{2...4}$ instead of k_n

There is a number of reasons why k_n coefficients are not used as function parameters, but derived from $m_{2...4}$ as above.

- At first, when $m_{2...4}$ coefficients are used as parameters, spiral arm passes through the points (r_0, φ_0) and $(r_{\text{end}}, \varphi_{\text{end}})$ regardless of $m_{2...4}$ values, and k_n coefficients are not independent from each other.

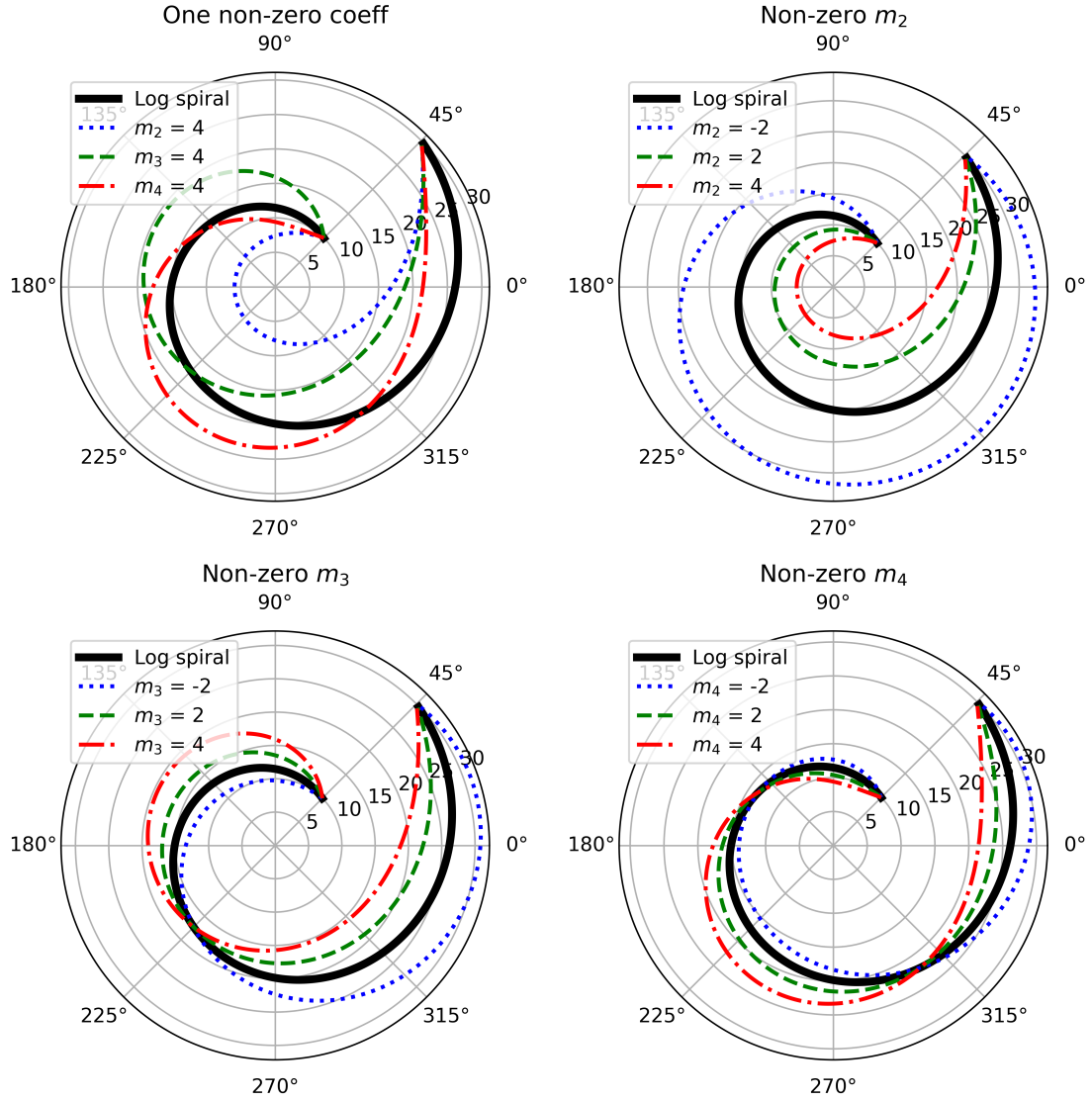


Figure 1: The influence of $m_{2...4}$ coefficients on the spiral arm shape. If all coefficients are equal to zero, the spiral is logarithmic.

- Parameters $m_{2...4}$ are closely tied with the Legendre polynomial expansion coefficients for the pitch angle (Eq. 4). Therefore, each of the coefficients $m_{2...4}$ represents the deviation of spiral arm shape from the logarithmic spiral, and coefficients make deviations mostly independent from each other because Legendre polynomials are orthogonal. If we would use k_n as function parameters, it would be some degree of degeneracy between them, because the increase of all k_n coefficients increases the pitch angle of the spiral arm, and for $k_{2...4}$ the increase is also stronger for the end of the arm.

The influence of $m_{2...4}$ coefficients on the spiral arm shape is shown on Fig. 1.

2.1.3 Pitch angle

Spiral arms in our model can have variable pitch angles. Pitch angle μ at the winding angle ψ is counted as following:

$$\mu(\psi) = \arctan \left(\sum_{n=1}^4 n k_n \psi^{n-1} / \psi_{\text{end}}^n \right). \quad (4)$$

Average pitch angle $\langle \mu \rangle$ over any ψ range can be found as an arctangent of the slope coefficient of the linear approximation of $\log r(\psi)$.

2.2 Distribution of the light along the arm: $I_{\parallel}$

Function $I_{\parallel}$ describes the light distribution along the ridge-line of the spiral arm, defined by $r(\psi)$. Because the spiral arm is a part of a disc, it's natural to assume that light distribution is exponential with radius at the most part of the spiral arm. To make the beginning and the ending of arm smooth, this exponential decrease is multiplied by the truncation function of ψ . The main part and the modification can be separated and be considered independently:

$$I_{\parallel}(r, \psi) = I_0^{\text{sp}} \times I_{r\parallel}(r) \times T(\psi) \quad (5)$$

$$I_{r\parallel}(r) = e^{-r \times i h_{\text{sp}}} \quad (6)$$

$$T_{\psi\parallel}(\psi) = s \left(\frac{\psi}{\psi_{\text{growth}}} \right) \times s \left(\frac{\psi_{\text{end}} - \psi}{\psi_{\text{end}} - \psi_{\text{cutoff}}} \right) \quad (7)$$

$$s(x) = \begin{cases} 0 & x < 0 \\ 3x^2 - 2x^3 & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases} \quad (8)$$

Above, I_0^{sp} is the projected surface brightness from the exponential part at the galaxy centre. Normally, spiral arm does not reach this brightness anywhere because the exponential decrease part of the arm begins from the off galaxy centre, as well as the arm as a whole. Such notation is chosen because it makes $I_{r\parallel}(r)$ have almost the same form as for exponential disc, with I_0^{sp} similar to the disc central surface brightness I_0 . The value $i h_{\text{sp}} = 1/h_{\text{sp}}$ is an inverse exponential scale of the spiral arm h_{sp} ¹. Both I_0^{sp} and $i h_{\text{sp}}$ are parameters of the function.

At ψ_{growth} , $I_{\psi\parallel}(\psi)$ stops increasing and becomes equal to 1, and therefore $I_{\parallel}(r, \psi)$ becomes purely exponential with radius (if $\psi_{\text{cutoff}} > \psi_{\text{growth}}$). At ψ_{cutoff} , $I_{\psi\parallel}(\psi)$ starts decreasing from 1, and $I_{\parallel}(r, \psi)$ overall decreases faster than exponentially with radius, reaching zero at the end. Instead of these angles, normalized values of them (as parts of total arm winding angle) are used as function parameters: $p_{\text{growth}} = \psi_{\text{growth}}/\psi_{\text{end}}$ and $p_{\text{cutoff}} = \psi_{\text{cutoff}}/\psi_{\text{end}}$, for the reasons of universality.

¹Despite exponential scale is widely used for discs description, we do not use it for spirals. Spiral arms sometimes have increasing or constant brightness towards higher r , and h_{sp} in these cases would be negative and/or have very large absolute value. Meanwhile, h_{sp} close to zero is usually unreasonable, so using $i h_{\text{sp}}$ as a parameter ensures that the range of reasonable parameters is continuous and compact.

2.3 Distribution of the light across the arm: $I_{\perp}$

Distribution of the light across the arm is defined by the function $I_{\perp}$ which is a simple symmetric Gaussian. We introduce $\rho = r - r(\psi)$ which is a radial distance from the point to the ridge-line of the spiral arm.

$$I_{\perp}(\rho, r) = \exp \left(-\ln(2) \times \left(\frac{\rho}{w_{\text{loc}}(\psi)} \right)^2 \right) \quad (9)$$

Here, local width $w_{\text{loc}}(r)$ is a HWHM² of the arm in a radial direction, which dependence of r is the following:

$$w_{\text{loc}}(r) = (w_{\text{zp}} + w_i r)/2 \quad (10)$$

Therefore, w_{zp} and w_i are linear coefficients defining width as a function of galactocentric radius.

3 Additional functions

3.1 Varieties of the function with bending

We implement functions which have one bending (two smooth parts, **SpiralArm1b**) and two bends (three smooth parts, **SpiralArm2b**). At the location(s) of bend(s), pitch angle changes abruptly. Formally speaking, $r(\psi)$ for these functions is defined piecewise. Over each smooth part, $r(\psi)$ is a polynomial function, but the coefficients are independent at each smooth part. For once-bent model, there are 2 polynomials of 3rd degree, and for twice-bent there are 3 polynomials of 2nd degree. The location(s) of one (or two) bends ($r_{\text{break}}, \varphi_{\text{break}}$) are also included into the list of function parameters. Obviously, adjacent smooth parts meet at the bending position. This is the only difference between the function varieties.

3.2 Fine-tuned profiles

As an extension to functions described before, we implement functions with extra parameters which allow one to add fine tuning to brightness distribution of spiral arms: add skewness of the profile, generalise Gaussian profile as Sérsic and add brightness dips.

These extensions for **SpiralArm**, **SpiralArm1b** and **SpiralArm2b** are called, respectively, **SpiralArmFull**, **SpiralArmFull1b** and **SpiralArmFull2b**. Now the properties of these extensions will be presented as a difference from original functions.

First, we describe the changes to $I_{\perp}$ (Equation 9):

$$I_{\perp}^{\text{in/out}}(\rho, \psi) = \exp \left(-\ln(2) \times \left(\frac{|\rho|}{w_{\text{loc}}^{\text{in/out}}(r)} \right)^{\frac{1}{n}} \right) \quad (11)$$

This function now takes the form of asymmetric Sérsic profile with an additional parameter n which defines the shape of the profile. It still has Gaussian as a specific case

²HWHM (half-width at half maximum) is a distance from the center to the location where the intensity equals half of peak intensity

if $n = 0.5$. One can note that, instead of b_n depending on n , the constant $\ln(2)$ stays in the exponent. This leads to $w_{\text{loc}}^{\text{in/out}}(\psi)$ being not an effective radius containing a half of spiral arm luminosity³ but a HWHM (just as for the baseline function). It is more convenient to use such definition in practice.

Indices ⁱⁿ and ^{out} denote, respectively, inner part of the arm relative to ridge-line of the arm ($\rho < 0$), and outer ($\rho > 0$), because inner and outer half-widths $w_{\text{loc}}^{\text{in}}$ and $w_{\text{loc}}^{\text{out}}$ are also different. Equation 10 takes the following form:

$$w_{\text{loc}}^{\text{in/out}}(r) = (w_{\text{zp}} + w_i r) \frac{1 \mp S_{\text{loc}}(r)}{2} \quad (12)$$

where $\mp$ means “−” for inner and “+” for outer. The local skewness $S_{\text{loc}}(r)$ is defined as following:

$$S_{\text{loc}}(r) = S_{\text{zp}} + S_i r \quad (13)$$

Instead of S_{zp} and S_i , function parameters S_0 and S_{end} are used, which define $S_{\text{loc}}(r_0)$ and $S_{\text{loc}}(r_{\text{end}})$ respectively.

Also, there is an extension to $I_{\parallel}$ (Eq. 5):

$$I_{\parallel}(r, \psi) = I_0^{\text{sp}} \times I_{r\parallel}(r) \times T(\psi) \times D(\psi) \quad (14)$$

where $D(\psi)$ is a function describing surface brightness dips in spiral arms based on Gaussian functions:

$$\begin{aligned} D(\psi) &= 1/[1 + \sum_i G(A_i, \psi_i^c, \psi_i^w)] \\ G(a, b, c) &= a \exp\left[-\frac{(x-b)^2}{2c^2}\right] \end{aligned} \quad (15)$$

Therefore, for each dip A_i, ψ_i^c, ψ_i^w are brightness dip depth, its location and width respectively. Our function supports up to 2 dips in a single arm, so we have additional parameters $A_1, \psi_1^c, \psi_1^w, A_2, \psi_2^c, \psi_2^w$.

3.3 Spurs

Finally, the function **SpiralSpur** is essentially a reduced **SpiralArm**. It has smaller number of parameters: m_3, m_4, ih_{sp} and w_{zp} are removed and the function is the same to **SpiralArm** with all these parameters are set zero. This simplified function is designed to fit smallest spiral features.

4 Model parameters list

³Actually, if b_n is taken from a traditional form of Sérsic function, the value that is considered to be an effective radius in common 2D Sérsic profile will encircle much more than a half of light, depending on n . For typical n of spiral arms in the range of $0.5 \dots 1$, this fraction equals nearly 80%.

4.1 Baseline function

For the baseline function `SpiralArm`, there is a following list of parameters:

Parameter	Name in IMFIT	Description and comments	Dimensions
	X0, Y0, PA, ell	Galactic center position and plane orientation (similar to other IMFIT components)	[px, px, deg, 1]
r_0, φ_0	r_0, phi_0	Spiral arm beginning position in polar coordinates, counted from the PA direction	[px, deg]
$r_{\text{end}}, \varphi_{\text{end}}$	r_end, phi_end	Spiral arm ending position in polar coordinates	[px, deg]
m_2, m_3, m_4	mu_a_2, mu_a_3, mu_a_4	Coefficients describing the deviation of spiral arm from pure logarithmic spiral shape (which is achieved when these coefficients are equal to zero). Normally, the absolute values of them are comparable or smaller than unity or a few	[1]
I_0^{sp}	I_0	Surface brightness projected to the galaxy center from the exponential decline part of the arm (similar to disc)	[1]
$p_{\text{growth}}, p_{\text{cutoff}}$	part_growth, part_cutoff	Parts of the arm (in terms of ψ) where growth from zero to exponential part, and then decline from exponential to zero occurs	[1]
ih_{sp}	ih_s	Inverse radial exponential scale of spiral arm	[1 / px]
w_{zp}, w_i	w_zp, w_i	Linear coefficients for arm width as a function of galactocentric radius	[px, 1]

4.2 Bendings

For functions with bendings (`SpiralArm1b` and `SpiralArm2b`, or `SpiralArmFull1b` and `SpiralArmFull2b`), we have the following parameters instead of m_2, m_3, m_4 :

Parameter	Name in IMFIT	Description and comments	Dimensions
m_2^a, m_3^a	mu_a_2, mu_a_3	Coefficients describing the deviation of the first spiral arm segment from pure logarithmic spiral shape	[1]
m_2^b, m_3^b	mu_b_2, mu_b_3	Coefficients describing the deviation of the second spiral arm segment from logarithmic spiral	[1]
m_2^c, m_3^c	mu_c_2, mu_c_3	Coefficients describing the deviation of the third spiral arm segment from logarithmic spiral (only for <code>SpiralArm2b</code>)	[1]
$r_{\text{break}}, \varphi_{\text{break}}$	r_break_1, phi_break_1	The location of the first bending of the spiral arm	[px, deg]
$r_{\text{break}}, \varphi_{\text{break}}$	r_break_2, phi_break_2	The location of the second bending of the spiral arm (only for <code>SpiralArm2b</code>)	[px, deg]

4.3 Fine tuning

Additional fine tuning coefficients (functions `SpiralArmFull`, `SpiralArmFull1b` and `SpiralArmFull2b`):

Parameter	Name in IMFIT	Description and comments	Dimensions
S_0, S_{end}	S_0, S_end	Perpendicular profile skewness at the beginning and at the end of spiral arm	[1]
n	n	Sérsic index of the perpendicular profile of the spiral arm	[1]
A_i	A_d_1, A_d_2	Brightness dip depths (max brightness decrease by a factor of $\exp[A_i]$; two independent dips are possible)	[1]
φ_i^c	phi_d_1, phi_d_2	Brightness dip locations (positional angle)	[deg]
φ_i^w	w_d_1, w_d_2	Brightness dip widths (positional angle)	[deg]